

GENERAL SPECIFICATION G3.21

Formwork

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G3.21 FORMWORK

G3.21.1 Scope

This section of the Specification covers the supply, erection and removal of formwork, the finishes to be attained and the remedial action to be taken to the finished concrete after removal of formwork.

G3.21.2 Submissions

When required by the S.O., the Contractor shall submit calculations and designs for formwork including of layout of panels, before fabrication commence.

G3.21.3 Materials

- (a) Formwork shall be constructed of timber, sheet metal or other approved material. The Contractor shall also furnish all struts, braces and ties to withstand the placing and vibrating of concrete, all construction loads and the effects of weather.
- (b) Form ties shall be the rod and cone or other approved proprietary type. They shall be designed so that no part remaining embedded in the concrete after formwork has been removed shall be nearer than 50 mm from the surface in the case of reinforced concrete and 150 mm in the case of unreinforced concrete.
- (c) Ties for use in water retaining structures 300 mm thick or less shall incorporate a diaphragm not less than 50 mm dia. welded to the midpoint of the tie, designed to prevent water passing along the tie.

G3.21.4 Design and Detailing

- (a) The Contractor shall be responsible for the adequacy and safety of formwork.
- (b) The design and construction of formwork shall take account of safety and of the surface finish required. The formwork shall be sufficiently rigid and tight to prevent loss of grout or mortar from the fresh concrete or the formation of fins or other blemishes on the concrete.
- (c) Formwork and its supports shall maintain their correct shape and profile so that the final concrete structure is within the limits of the specified dimensional tolerances. They shall be designed to withstand the worst combination of self-weight, reinforcement weight, wet concrete weight, concrete pressure, construction and weather loads, together with all incidental dynamic effects caused by placing, vibrating and compacting the concrete. Formwork shall not be tied to or supported by the reinforcement.
- (d) On formwork to external faces which will be permanently exposed, all horizontal and vertical formwork joints shall be so arranged that joint lines shall form a uniform pattern on the face of the concrete. Where the Contractor proposes to make up the formwork from standard sized manufactured formwork panels, the size of such panels shall be approved by the S.O. before they are used in the construction of the work. The finished appearance of the entire elevation of the structure and adjoining structures shall be considered when planning the pattern of joint lines caused by formwork and by construction joints to ensure continuity of horizontal and vertical lines.

G3.21.5 Formed Surfaces - Class of Finish

- (a) Finishes to formed surfaces of concrete shall be classified as F1, F2 or F3, or such special finish as may be specified.
- (b) Formwork for Class F1 finish shall be constructed of timber, sheet metal or any suitable materials which will prevent loss of grout when the concrete is vibrated. Surfaces subsequently to be rendered, plastered or tiled shall be adequately scabbled or roughened as soon as the formwork is removed to reduce the irregularities to not more than half the thickness of such rendering, plastering or bedding for tiles and to provide a satisfactory key.
- (c) Class F2. The irregularities in the finish shall be no greater than those obtained from the use of wrought thicknesses square edged boards arranged in a uniform pattern. The finish is intended to be left as struck but imperfections such as fins and surface discolouration shall be made good.
- (d) Class F3. The resulting finish shall be smooth and of uniform texture and appearance. The Formwork lining shall leave no stain on the concrete and shall be so joined and fixed to its back that it imparts no blemishes. It shall be of the same type and obtained from only one source throughout any one structure. The Contractor shall make good and imperfections in the finish. The ties shall be positioned only in rebates or in other positions as described in the contract or as accepted by the S.O.
- (e) Unless otherwise specified or shown on the Drawings, formed concrete finishes F1, F2 or F3 shall be used at the following locations:
 - F1 a) Formwork to mass concrete which is to be buried.
 - F2 a) Formwork to buried external reinforced concrete surfaces from 500 mm below final ground level to bottom of concrete surface;
 - b) Formwork to inside of external chambers (valve chambers, flowmeters and the like) which do not come into contact with the potable water stream;
 - c) Internal surfaces to general basements; and
 - d) Structural members of buildings including all surfaces of columns, beams and slabs above ground level which will subsequently be plastered, hidden above suspended ceilings or clad in some other architectural finish.
 - F3 a) Exposed reinforced concrete surfaces from 500 mm below final ground level to top of concrete surface;
 - b) Internal surfaces (including soffits) of process tanks;
 - c) Exposed reinforced concrete surfaces in areas of general personnel access in process plant areas (e.g. filter galleries) including soffits; and
 - d) Structural members of buildings (excluding basements) but including all surfaces of columns, beams and slabs above ground level which will not subsequently be plastered, hidden above suspended ceilings or clad in some other architectural finish.

G3.21.6 Erection of Formwork

- (a) All formwork shall be soundly constructed, firmly supported, adequately struted, braced and tied to withstand the placing and vibrating of concrete, construction loads and the effects of weather. Formwork shall not be tied to or supported by reinforcement.
- (b) Faces of formwork in contact with concrete shall be free from adhering foreign matter, projecting nails and the like, splits or other defects, and all formwork shall be clean and free from standing water, dirt, shavings, chippings or other foreign matter. Joints shall be sufficiently watertight to prevent the escape of mortar or the formation of fins or other blemishes on the face of the concrete.
- (c) All exposed exterior angles on the finished concrete of 90 degrees or less shall be given 20 mm by 20 mm chamfers.
- (d) Formwork in contact with the concrete shall be treated with a suitable non-staining mould oil prior to reinforcement and concrete placement to prevent adherence of the concrete. Care shall be taken to prevent the oil from coming in contact with reinforcement or with concrete at construction joints. Surface retarding agents shall not be used unless specified.
- (e) Where ties are built into the concrete for the purpose of supporting formwork, the whole or part of any such supports shall be capable of removal so that no part remaining embedded in the concrete shall be nearer than 50 mm from the surface in the case of reinforced concrete and 150 mm in the case of unreinforced concrete. Holes left after removal of such supports shall be neatly filled with well rammed dry pack mortar.
- (f) Openings for inspection of the inside of the formwork, for the removal of water used for washing down and for placing concrete shall be provided and so formed as to be easily closed before or during placing concrete. Before placing concrete all bolts, pipes or conduits or any other fixture which are to be built in shall be held fast by fixing to the formwork or otherwise. Holes shall not be cut in any concrete without prior approval of the S.O..
- (g) Concrete shall not be placed against formwork until the S.O. has inspected and accepted the condition of the formwork. The Contractor shall give the S.O. a minimum of four working hours' notice to undertake such inspections.

G3.21.7 Removal of Formwork

- (a) Formwork shall be so designed as to permit easy removal without resorting to hammering or levering against the surface of the concrete.
- (b) Formwork supporting cast-in-situ concrete shall not be removed unless:
 - (i) Such removal will not result in unacceptable deflections due to shrinkage, creep, etc.;
 - (ii) Such removal has been accounted for in the sequence of construction that has been previously agreed upon between the Contractor and the S.O.; and
 - (iii) The cube strength of the cast concrete is 10 N/mm² or twice the stress to which it will then be subjected, whichever is higher.

- (c) In the absence of the cube test result and stress calculations, the minimum periods before striking of the formwork for Ordinary Portland Cement concrete, as given in table hereunder, may be accepted by the S.O. provided that:
- (i) The requirements of the preceding paragraph's subclause (i) and (ii) are satisfied; and
 - (ii) The Contractor accepts the responsibility for the strength of the concrete as required by the preceding paragraph's subclause (iii).

Location	Minimum Period for Ordinary Portland Cement Concrete
Sides of columns, walls and beams	1 day
Soffits of slabs (props left under)	3 days
Soffits of beams (props left under)	7 days
Props to slabs	7 days
Props to beams	14 days

- (d) In any event, no formwork or support shall be struck or moved without the S.O.'s consent in writing and shall be done only in the presence of the S.O.. The permission of the S.O. shall not relieve the Contractor of any of his responsibilities for the structure.
- (e) Formwork shall be removed by gradual easing without shock or vibration and in a manner and order approved by the S.O.. The Contractor shall be responsible for any injury to the work and any consequential damage caused by or arising from the method adopted for the striking or moving of the formwork or supports.
- (f) Immediately after the removal of the formwork, it is the Contractor's responsibility to notify the S.O. for inspection and any remedial treatment required by the S.O. shall be carried out without delay. Any remedial treatment which has been carried out before the S.O.'s inspection and approval shall render the member liable for rejection.
- (g) If the Contractor has been permitted to use rapid hardening Portland cement other than where specified, to expedite, for his own advantage, the progress of the Works, it shall be understood that notwithstanding the advantage in time which is gained by the earlier stripping of shuttering, no extra payments shall be made by the Board for the use of these alternative cements.
- (h) After removal of formwork no surface texture or touching up shall be attempted until the work has been inspected by the S.O.. If, in his opinion, any defect cannot be made good by touching up he may direct the Contractor to replace such work at the Contractor's expense.

- (i) Upon removal of the shuttering and with the approval of the S.O. honeycombed surfaces shall be made good immediately by a method approved by the S.O. and superficial water and air holes shall be filled in. Unless instructed to the contrary, the face of exposed concrete placed against shuttering shall be rubbed down with a carborundum stone immediately upon removal of the shuttering to remove fins or other irregularities. The face of concrete for which shuttering is not provided, other than slabs, shall be floated to give a finish equal to that of the rubbed down face where shuttering is provided. No cement wash, mortar or paint may be applied to any concrete surface without the permission of the S.O.. Generally concrete finishes as struck from the shuttering are required.

G3.21.8 Dry Pack Mortar

- (a) Dry pack mortar for filling holes and repairing surface blemishes shall be made from one part by weight of cement and three parts fine aggregate passing a 1 mm sieve. The colour of the mortar shall match that of the surrounding concrete. The mortar shall be mixed with only sufficient water to make the materials stick together when being moulded in the hands.
- (b) The dry pack material shall then be placed and packed in layers having a thickness not greater than 15 mm. The compaction shall be carried out by use of hardwood stick and hammer and shall extend over the full area of the layer, particular care being taken to compact the dry pack against the sides of the hole. After compaction, the surface of each layer shall be scratched before further loose material is added. Holes shall not be over filled and the surface shall be finished by laying the hardwood block against the dry pack fill and striking the block several times. Steel finishing tools shall not be used and water shall not be added to facilitate finishing.

G3.21.9 Defects in Formed Surfaces

- (a) Workmanship in formwork and concreting shall be such that concrete shall require no making good, surfaces being perfectly compacted and smooth.
- (b) Any surface blemishes shall be repaired by the Contractor, at his expense, to the satisfaction of the S.O. immediately after removal of formwork. Any remedial treatment to concrete surfaces shall be agreed with the S.O..
- (c) No rendering of finished surfaces to make good shall be allowed.
- (d) Any concrete the surface of which is found to have been treated before inspection by the S.O. shall be rejected.
- (e) Remedial measures may include, but shall not be limited to, the following:
 - (i) Holes left by the formwork supports shall be thoroughly cleaned out to remove all loose material and the sides shall be roughened, if necessary, to ensure a satisfactory bond. They shall then be filled with an approved proprietary epoxy resin mortar or with dry pack mortar if agreed by the S.O.;
 - (ii) Fins, pinhole bubbles, surface discolouration and minor defects may be rubbed down with sacking and cement immediately after the formwork is removed;
 - (iii) Abrupt and gradual irregularities may be rubbed down with carborundum and water after the concrete has been fully cured; and

- (iv) Small defects and minor honeycombing shall be chipped out normal to the face of the concrete to a depth of at least 25 mm and filled with an approved proprietary epoxy resin mortar or with dry pack mortar if agreed by the S.O..
- (f) Where deeper or more extensive defects occur, the Contractor shall obtain the approval of the S.O. to the methods of repair proposed which may include, but shall not be limited to, cutting out to a depth of 25 mm with a diamond saw to give a regular edge to the repair and further chipping to form a hole with dovetail shape to sound concrete or to a total depth of 75 mm whichever is the greater. If reinforcing steel is exposed, the concrete shall be removed to a depth of 25 mm beyond the back side of the reinforcement. Steel mesh reinforcement shall then be sprung into the dovetail. The void shall be refilled with concrete or suitable approved epoxy resin mortar.
- (g) The Contractor shall thoroughly clean any hole or defective area that is to be filled and where the surface has been damaged the Contractor shall break out any loose, broken or cracked concrete or aggregate.
- (h) Where the remedial work is to be carried out using dry pack mortar or concrete, the concrete surrounding the hole shall be thoroughly soaked after which the surface shall be dried so as to leave a small amount of free water on the surface. The surface shall then be dusted lightly with cement by means of a small dry brush until the whole surface that will come into contact with the dry pack mortar has been covered and darkened by absorption of the free water by the cement. Any dry cement in the hole shall be removed.
- (i) Dry pack mortar shall be mixed and placed as specified.
- (j) Where concrete is to be used, the concrete mix shall be approved by the S.O. and shall be placed and compacted into the hole, using formwork as necessary.
- (k) Where the remedial work is to be carried out using epoxy resin mortar or other specialist material, the surface of the cleaned hole shall be prepared and primed and the repair material placed, compacted and finished in accordance with the manufacturer's instructions.
- (l) Where in the opinion of the S.O., the defect is too extensive to permit satisfactory repair, either from the point of view of structural integrity or appearance, the concrete containing the defect shall be broken out and replaced.

G3.21.10 Inspection and Testing

- (a) Before concrete is placed against any formwork, the formwork shall be inspected and approved by the Contractor and offered for inspection by the S.O..

END OF SECTION